

## Summary

This video provides a comprehensive lecture on the Theory of Computation, focusing on the construction of Regular Expressions (RE) and their relationship with Deterministic Finite Automata (DFA). The analysis covers positional symbol constraints, string length requirements (including modular arithmetic), and start/end symbol logic. A central takeaway is the “AD Rule” for determining DFA states based on right-end positional constraints. This document serves as a structured technical guide for constructing REs that satisfy specific linguistic patterns relevant for competitive examinations such as GATE 2026.

### 1. Positional Symbol Constraints

Regular expressions can enforce symbol placement at specific indices relative to the left (start) or right (end) of a string over the alphabet  $\{a, b\}$ .

#### 1.1 Left-End Constraints

To fix a symbol at the  $n$ -th position from the left, the preceding  $(n - 1)$  positions must allow any symbol from the alphabet, represented as  $(a+b)$ .

Examples:

- 2nd symbol is 'a'

Regular Expression:  $(a+b)a(a+b)^*$

Explanation: The first position is unrestricted; the second position is fixed as 'a'.

Expanded form:  $(aa + ba)(a+b)^*$

- 3rd symbol is 'b'

Regular Expression:  $(a+b)(a+b)b(a+b)^*$  or  $(a+b)^2 b (a+b)^*$

Explanation: The first two positions are unrestricted; the third is fixed as 'b'.

Possible prefixes: aab, abb, bab, bbb.

- 4th symbol is 'a'

Regular Expression:  $(a+b)^3 a (a+b)^*$

In general, for the  $n$ -th symbol from the left fixed as  $x$ :

$(a+b)^{(n-1)} x (a+b)^*$

#### 1.2 Right-End Constraints

To fix a symbol at the  $n$ -th position from the right, the suffix must be explicitly defined, and the prefix remains arbitrary.

Examples:

- 2nd symbol from right is 'a'

Regular Expression:  $(a+b)^* a (a+b)$

Interpretation: The string ends in either "aa" or "ab".

- 3rd symbol from right is 'b'

Regular Expression:  $(a+b)^* b (a+b)^2$

Possible endings: baa, bab, bba, bbb.

- 4th symbol from right is 'a'

Regular Expression:  $(a+b)^* a (a+b)^3$

In general, for the n-th symbol from the right fixed as x:

$(a+b)^* x (a+b)^{n-1}$

## 2. Theoretical Optimization: The AD Rule

A structural relationship exists between right-end positional constraints and minimal DFA size.

### 2.1 State Calculation Logic

If the n-th symbol from the right is fixed, then:

Number of states in the minimal DFA =  $2^n$

This arises because the DFA must remember the last n symbols read, and there are  $2^n$  possible combinations over {a, b}.

### 2.2 Application Example (GATE 2017)

Language: 2nd symbol from right is 'b'

Regular Expression:  $(a+b)^* b (a+b)$

Here  $n = 2$

Minimal DFA states =  $2^2 = 4$

Note: For left-end positional constraints, state complexity generally follows an  $(n+1)$  pattern because the DFA tracks how many symbols have been read until the fixed position.

## 3. String Length Constraints

Length-based constraints may require exact, bounded, or modular conditions.

### 3.1 Basic Length Constraints

| Constraint Type | Regular Expression Logic |
|-----------------|--------------------------|
|-----------------|--------------------------|

|                   |                                                                    |
|-------------------|--------------------------------------------------------------------|
| Exactly length 2  | $(a+b)(a+b)$ or $(a+b)^2$                                          |
| At most length 2  | $\epsilon + (a+b) + (a+b)^2$ or $(\epsilon + a+b)(\epsilon + a+b)$ |
| At least length 2 | $(a+b)^2 (a+b)^*$                                                  |

### 3.2 Variations of “At Least Length 2”

Multiple equivalent constructions exist for length  $\geq 2$ :

1.  $(a+b)(a+b)(a+b)^*$
2.  $(a+b)^2 (a+b)^*$
3.  $(a+b)^* (a+b)^2$
4.  $(a+b)(a+b)^+ (a+b)$
5.  $(aa + ab + ba + bb)(a+b)^*$

Each form ensures at least two symbols before arbitrary continuation.

### 3.3 Modular Length Constraints

For strings whose length satisfies  $k \bmod n$ :

General Pattern:

$((a+b)^n)^* (a+b)^k$

Examples:

| Length Condition     | Regular Expression                   |
|----------------------|--------------------------------------|
| Even ( $0 \bmod 2$ ) | $((a+b)^2)^*$                        |
| Odd ( $1 \bmod 2$ )  | $((a+b)^2)(a+b)$ or $(a+b)((a+b)^2)$ |
| $2 \bmod 3$          | $((a+b)^3)^*(a+b)^2$                 |
| $3 \bmod 4$          | $((a+b)^4)^*(a+b)^3$                 |
| $2 \bmod 5$          | $((a+b)^5)^*(a+b)^2$                 |

These patterns rely on grouping into blocks of size  $n$  and appending a remainder block of size  $k$ .

## 4. Start and End Symbol Logic

Regular expressions can enforce constraints relating the first and last symbols.

- Starts and ends with 'a'

Regular Expression:  $a(a+b)^*a + a$

The single 'a' is included since it satisfies both conditions.

- Starts and ends with same symbol

Regular Expression:

$(a(a+b)^*a + a) + (b(a+b)^*b + b)$

- Starts and ends with different symbols

Regular Expression:

$a(a+b)^*b + b(a+b)^*a$

Single symbols are excluded because they cannot satisfy different start and end simultaneously.

## 5. Combined Constraints: Start and Length

When combining start symbol constraints with length requirements, the fixed start symbol must be counted toward total length.

- Starts with 'a' and even length

Regular Expression:  $a(a+b)((a+b)^2)^*$

Explanation:

'a' contributes length 1

$(a+b)$  contributes length 1

Total base = 2 (even)

Additional growth in blocks of 2 maintains even parity.

- Starts with 'b' and odd length

Regular Expression:  $b((a+b)^2)^*$

Explanation:

'b' contributes length 1 (odd)

Repeating blocks of 2 preserve odd parity.

## 6. Advanced Numerical Constraints: $a^m b^n$

Consider languages of the form  $L = \{ a^m b^n \}$  with parity constraints on  $(m+n)$ .

### 6.1 Case: $(m + n)$ is Even

Occurs in two scenarios:

1. m even and n even:

$(aa)(bb)$

2. m odd and n odd:

$(aa)^*a(bb)^*b$

Combined Regular Expression:

$(aa)(bb) + (aa)^*a(bb)^*b$

6.2 Case:  $(m + n)$  is Odd

Occurs in two scenarios:

1.  $m$  even and  $n$  odd:

$(aa)^*(bb)^*b$

2.  $m$  odd and  $n$  even:

$(aa)a(bb)$

Combined Regular Expression:

$(aa)^*(bb)^*b + (aa)a(bb)$

These constructions ensure that all 'a's precede all 'b's while satisfying parity conditions.